

# CS 105: Department Introductory Course on Discrete Structures

Instructor : S. Akshay

Graph theory  
Perfect Matchings and Hall's theorem

Lecture 22  
Oct 27 2025

## Topic 4: Graph theory

### Basic definitions and concepts

- ▶ Basics: graphs, paths, cycles, walks, trails, ...
- ▶ Cliques and independent sets.
- ▶ Graph representations, isomorphisms and automorphisms.
- ▶ Matchings: perfect, maximal and maximum.

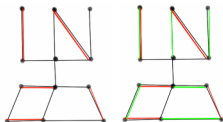
### Characterizations

1. **Eulerian graphs:** Using degrees of vertices.
2. **Bipartite graphs:** Using odd length cycles.
3. **Connected components:** Using cycles.
4. **Maximum matchings:** Using augmenting paths.
5. Today: **Perfect matchings in bipartite graphs**

## Recap: Matchings

### Definitions

- ▶ **Matching**: set of edges with no shared end-points.
- ▶ Vertices incident to edges in matching are **saturated**. Others are **unsaturated**.
- ▶ **Perfect matching**: saturates every vertex in graph.
- ▶ **Maximum matching**: matching of maximum size (# edges).
- ▶ **Maximal matching**: cannot be enlarged by adding an edge.
- ▶  **$M$ -alternating path**: alternates between edges in/out of  $M$ .
- ▶  **$M$ -augmenting path**:  $M$ -alternating path with endpoints unsaturated by  $M$ .



## Recap: Characterizing maximum matchings

### Theorem (Berge'57)

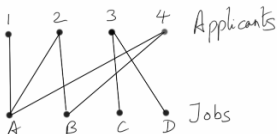
A matching  $M$  in  $G$  is a maximum matching iff  $G$  has no  $M$ -augmenting path.

Proof:

- ▶ ( $\implies$ ) Contrapositive: If  $G$  has  $M$ -augmenting path it can't be maximum!
- ▶ ( $\impliedby$ ) For reverse also, we show contrapositive. i.e., if  $\exists$  matching  $M'$  larger than  $M$ , we will construct an  $M$ -augmenting path.
- ▶ Let  $F = M \triangle M'$ . By Lemma,  $F$  has only paths and even cycles with equal no. of edges from  $M$  and  $M'$ .
- ▶ Since  $|M'| > |M|$  it must have a component with more edges in  $M'$  than  $M$ .
- ▶ This component can only be a path that starts and ends with an edge of  $M'$ ; i.e., it is an  $M$ -augmenting path in  $G$ . □

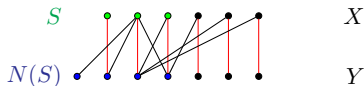
## Perfect matchings in bipartite graphs

- **Pop-Quiz!:** If there are  $n$  women and  $n$  men, and each woman is compatible with exactly  $k$  men and each man compatible with exactly  $k$  women, can they be perfectly matched?
- If there are  $m$  jobs and  $n$  applicants, when can we find a perfect matching where all  $m$  jobs are saturated?



## Perfect matchings in bipartite graphs

- ▶ Consider a bipartite graph with  $X, Y$  as partitions.
- ▶ If a matching  $M$  saturates  $X$ , then for every  $S \subseteq X$ , their neighbours in  $Y$ ,  $N(S)$  must exist at least as many as  $|S|$ .
- ▶ That is,  $\forall S \subseteq X, |N(S)| \geq |S|$  (**Hall's Condition**). This is a necessary condition, is it sufficient?



### Theorem (Hall'35)

A bipartite graph  $G$  with bipartitions  $X, Y$  has a matching that saturates  $X$  iff for all  $S \subseteq X$ ,  $|N(S)| \geq |S|$ .

## Characterizing perfect matchings in bipartite graphs

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- ▶ For  $v \in V$ , its neighbour-set  $N(v) = \{u \in V \mid (u, v) \in E\}$ .
- ▶ For  $S \subseteq V$ ,  $N(S) = \{u \in V \mid (u, v) \in E \text{ for some } v \in S\}$ .

## Characterizing perfect matchings in bipartite graphs

### Theorem (Hall'35)

A bipartite graph  $G$  with bipartitions  $X, Y$  has a matching that saturates  $X$  iff for all  $S \subseteq X$ ,  $|N(S)| \geq |S|$ .

Proof: ( $\implies$ ) is straightforward:

- ▶ Let  $M$  be a matching.
- ▶ Then for any  $S \subseteq X$ , each vertex of  $S$  is matched to a distinct vertex in  $N(S)$
- ▶ So  $|N(S)| \geq |S|$ .



## Characterizing perfect matchings in bipartite graphs

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Proof: ( $\Leftarrow$ )

- ▶ **Converse:** If  $\forall S \subseteq X, |N(S)| \geq |S|$ , then  $G$  has a matching that saturates  $X$ .
- ▶ **Contrapositive:** If  $G$  does not have any matching that saturates  $X$ , then there must exist  $S \subseteq X, |N(S)| < |S|$ .
- ▶ If  $G$  does not have any matching that saturates  $X$ , then surely any maximum matching of  $G$  does not saturate  $X$ .
- ▶ Let  $M$  be such a maximum matching. Then, we will construct  $S \subseteq X$  s.t.  $|N(S)| < |S|$ .

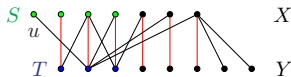
## Characterizing perfect matchings in bipartite graphs

### Theorem (Hall'35)

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Proof: ( $\Leftarrow$ ) Thus, starting from a maximum matching  $M$  which does not saturate  $X$ , we construct  $S \subseteq X, |N(S)| < |S|$ .

- ▶ Let  $u \in X$  be any unsaturated vertex of  $M$ .
- ▶ Consider  $V_u$ , set of vertices reachable from  $u$  by  $M$ -alternating paths in  $G$ .
- ▶ Let  $S = V_u \cap X$  and  $T = V_u \cap Y$ .



Claim:  $M$  matches  $T$  with  $S \setminus \{u\}$  and  $|N(S)| = |T|$ .

- ▶ Every vertex of  $S \setminus \{u\}$  has an edge in  $M$  to a vertex in  $T$ .
- ▶ Every vertex of  $T$  extends via  $M$  to a unique vertex of  $S$  (no  $M$ -aug path).
- ▶ Thus, there is a bijection between  $T$  and  $S \setminus \{u\}$ .

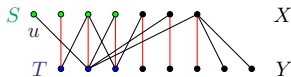
# Characterizing perfect matchings in bipartite graphs

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Claim:  $M$  matches  $T$  with  $S \setminus \{u\}$  and  $|N(S)| = |T|$ .

- ▶  $T \subseteq N(S)$  (from  $T$  any  $M$ -alternating path will reach  $S$ ).
- ▶ Conversely, if  $v \in S$  has edge to  $y \in Y \setminus T$ , then path from  $u$  to  $v$  via  $M$  to  $y$  is an  $M$ -alternating path, implies  $y \in T$ .

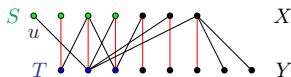
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Claim:  $M$  matches  $T$  with  $S \setminus \{u\}$  and  $|N(S)| = |T|$ .

Thus,  $|N(S)| = |T| = |S| - 1 < |S|$

□.

## Applications of Hall's condition

### Theorem (Hall'35)

A bipartite graph  $G$  with bipartitions  $X, Y$  has a matching that saturates  $X$  iff for all  $S \subseteq X$ ,  $|N(S)| \geq |S|$ .

### The Marriage Theorem (1917)

- ▶ In a group of  $n$  women and  $n$  men, if every woman is compatible with  $k$  men and every man compatible with  $k$  women, then a perfect matching must exist!
- ▶ For  $k > 0$ , every  **$k$ -regular** bipartite graph (i.e, **every vertex has degree exactly  $k$** ) has a perfect matching. **Ex. Prove this!**
- ▶ If  $G$  is a  $k$ -regular  $X, Y$  bipartite graph, then  $|X| = |Y|$ .
- ▶ If a matching saturates  $X$  then it saturates  $Y$ .
- ▶ Can you now verify Hall's condition?

## Applications of Hall's condition

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- ▶ If  $G$  is a  $k$ -regular  $X, Y$  bipartite graph, then  $|X| = |Y|$ .
- ▶ If a matching saturates  $X$  then it saturates  $Y$ .
  - ▶ Let  $S \subseteq X$ . Let  $m = \#$  edges from  $S$  to  $N(S)$ .
  - ▶ Since  $G$  is  $k$ -regular,  $m = k|S|$ . And since they touch  $N(S)$ ,  $m \leq k|N(S)|$ . Hence  $k|S| \leq k|N(S)|$ ,  $k > 0$  completes proof. □

## Let's play a game

### A two player game on a graph

1. Given a graph  $G$ , two players will alternatively choose distinct vertices.
2. One player starts by choosing any vertex.
3. Subsequent move must be adjacent to preceding choice (of other player).
4. Last player who can move wins.

Two volunteers please. Who wants to start?

Application 2: Theorem (H.W: Qn 3.1.18 from Douglas West)

If  $G$  has a perfect matching, then player 2 has a winning strategy; otherwise, player 1 has a winning strategy.